

OSI Model :-

Why OSI model created?

The OSI model is created to allow devices to communicate irrespective of the vendors.

Its goals were:

- (i) standardization
- (ii) Interoperability
- (iii) make devices from different companies work together.

Why Layers?

Suppose you're building a house you don't ask one person to do everything instead each person do their individual work.

Networking works exactly the same way.

⇒ Responsibilities divided into layers.

Each layer has one job.

The OSI Model

There are 7 layers

- 4) Application layer All
- 6) presentation layer people
- 5) session layer sleep
- 4) Transport layer through
- 3) Network layer Networking
- 2) Data Link layer Don't
- 1) physical pers

But we don't actually use OSI model.

* OSI model is mainly for understanding networking

* real networks use the TCP/IP

original TCP/IP model :

4 layers

- 1) Application
- 2) Transport
- 3) Internet
- 4) Link

Modern TCP/IP Model (Hybrid)

5 layers

1) Application

2) Transport

3) Network

4) Data Link

5) physical

Layer 1 - physical layer

* Move bits through some physical layer medium.

* It doesn't know

→ IP

→ MAC

→ websites

* It only knows 0s & 1s

These bits are transmitted as

→ electrical signals (Ethernet copper)

→ light (fiber)

→ radio waves (Wi-Fi)

→ cellular signals (4G/5G)

→ satellite signals (Starlink)

Devices at Layer 1:

Hub, cables, connectors, repeaters

Layer 2 - Data Link Layer

Now the data has reached another device.

a) How does it know which nearby device should receive it?

Layer 2 introduces MAC addresses.

The switch looks only at MAC addresses.

Main job of Layer 2

- * MAC addressing

- * Framing

- * Error detection

- * ~~Can~~ Communicate between directly connected devices.

What is a Frame?

At Layer 2 data becomes a Frame

Frame

Destination MAC

Source MAC

payload

CRC

payload is the actual data being transferred.

Why Directly connected device?

Suppose $PC \rightarrow Router \rightarrow Router \rightarrow Server$

* Between PC to Router

* Layer 2 uses 'one frame'

* When the router forwards it, that frame is destroyed.

* A completely new frame is created for Router $\rightarrow$ Router.

* So Layer 2 exists only between neighbouring devices.

* It changes at every hop.

Layer 3 - Networking Layer

* This Layer introduces IP addresses

* Unlike MAC addresses, IP addresses work across many networks.

* Routers don't care about MAC addresses from another network.

* They care about IP addresses.

Main Job :

- * Logical addressing
- * Routing
- * Best path selection

Devices

Routers operate here

Layer 4 - Transport layer

This layer is responsible for communication between two applications.

Two major protocols

* TCP

* UDP

Layer 5 - Application Layer in TCP/IP

This is the layer user interacts with.

Eg: web browser, whatsapp, Gmail, etc.

Protocols :-

HTTP, HTTPS, FTP, SSH, Telnet

Encapsulation :

One of the most important concepts in networking is that each layer adds its own information to the data.

Message $\Rightarrow$ Hello

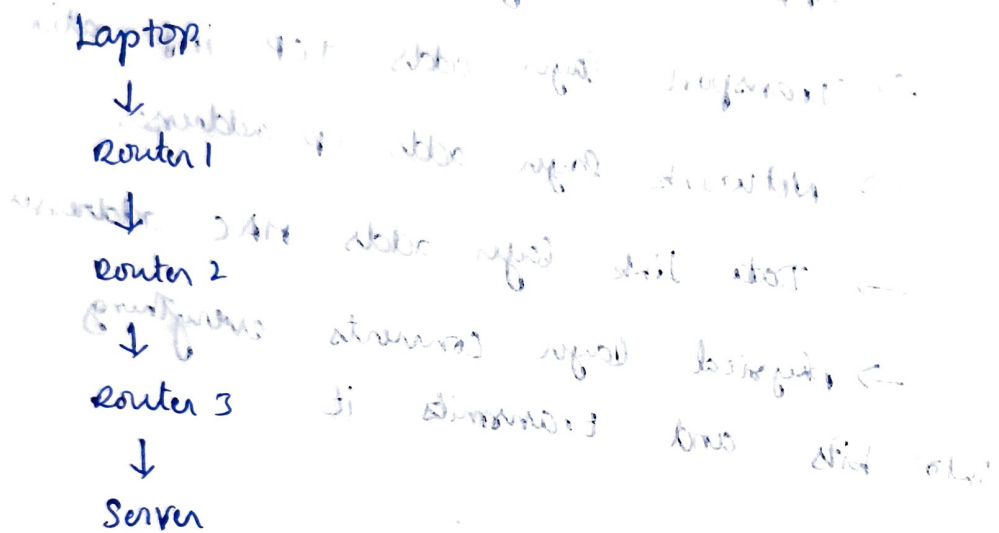
- $\rightarrow$ Application layer creates the data
- $\rightarrow$ Transport layer adds TCP information
- $\rightarrow$ Network layer adds IP addresses.
- $\rightarrow$ Data Link layer adds MAC addresses.
- $\rightarrow$ physical layer converts everything into bits and transmits it.

Data units at each layer :

Layer	Data Unit	Device
Application	Data	web servers, browser, ...
Transport	segment (TCP) / Datagram (UDP)	TCP / UDP (logical protocols)
Network	packet	Router
Data Link	Frame	switches, bridges
physical	bits	Hub, cables

Why Layer 2 changes but Layer 3 doesn't
This is one of the most important.

Suppose you send data from your laptop to a web server.



At each hop:-

* The Layer 2 frame is removed & replaced with a new one because each link has different neighbouring devices (and may use different technologies like ethernet, Wi-Fi, etc., PPP)

* The Layer 3 IP packet stays the same from the original sender to the final destination (ignoring special cases like NAT)

Layer 2 information changes at every hop while Layer 3 information is preserved end to end.

What is hop?

* One device (usually a router) forwarding your packet to the next device on its journey.

* Each router forwarding your packet is one hop.

* Every router increases the hop count by one.

Router $\rightarrow$ Router is a hop.